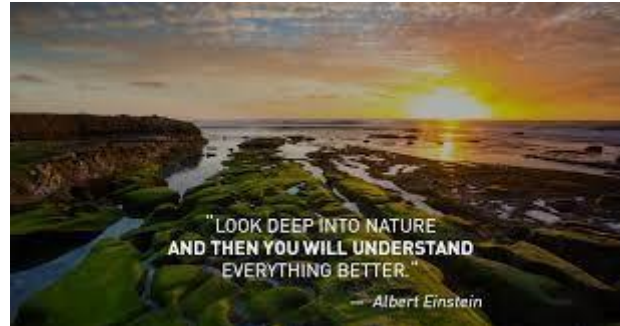


DESIGN THINKING

An insight of design:

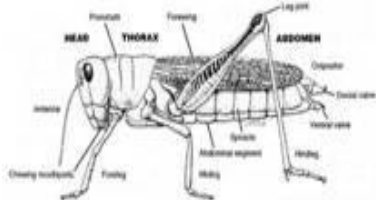
Nature Vs human



Plants, animals and human beings are creation of nature and one of the theories of evolution suggests that life forms began simply and then became more complex. Such a theory proposes that human beings are possibly a highly evolved creation with the ability to understand the mysteries and mechanisms of nature.

Designs inspired by nature:

Nature made	Man made
 Aves	 Flying suit
 Crane bird	 Crane



Grasshopper



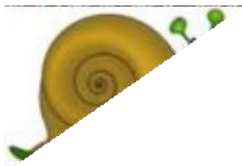
Grasshopper structure



Kangaroo pouch



Baby pouch



Snail



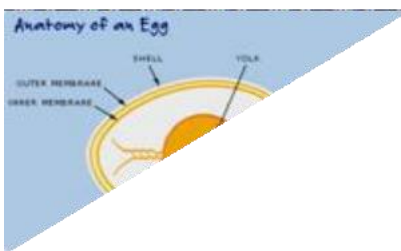
Cd track



Spiral climber



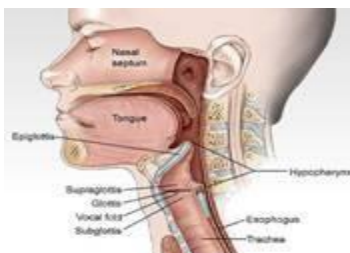
Stair case



Egg



Medicine capsules



Human anatomy



Pipes



Butter fly



Industrial pipe lines



Spider net



Fishing net



Basket ball net



Soccer ball net



Pelican



Pouch bags



Sapling



Leaf spring



Joints of Finger (Flexing)



Automobile



Closed Fist (Bending)



Elbow joints



Machine part joints



Blister on skin



Medicine pack



Pen drives pack



Lion fur



Crocodile skin



wrapped baby



Scales



Wrapped suitcase

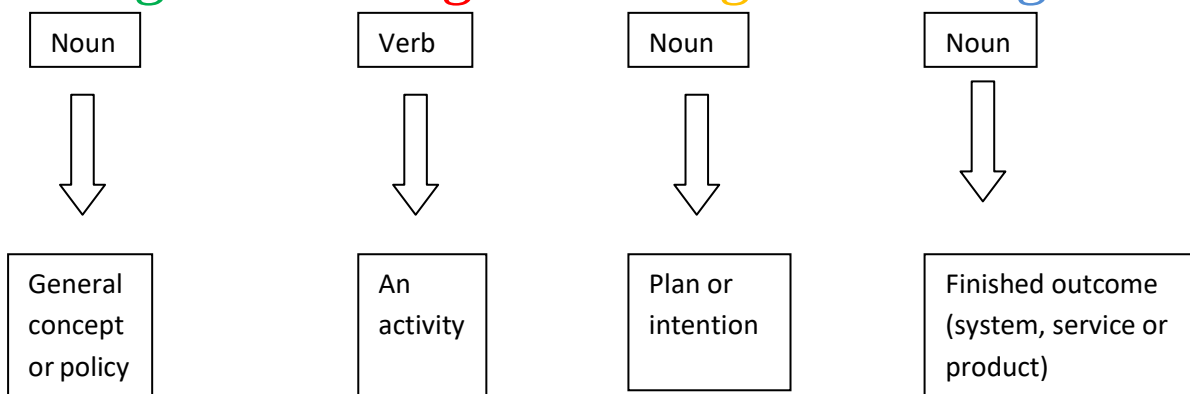
Definition of design:

- ❖ “The future is best found in the opportunities that go un noticed in the present”—**peter drucker** (Father of Management Thinking)
- ❖ ‘Everything that needs to be said has already been said. But since no one was listening ,everything must be said again’

According to Dictionary Design means

“A plan or drawing produced to show the look, the function or working of a building, garment, or other object before it is made”

Design is to design the design of a design



➤ Design(Noun):

Blue print of something-a plan for change

Undesirable situation (present) + Implemented plan= Desirable situation (future)

- Whether a situation is undesirable and what aspects are desirable depends on matter of perception
- Whose perception, where it is perceived, and when it is perceived plays important role.

➤ **Design(verb):**

- Understanding & solving a problem:: termed as Designing
- **Problem understanding:** process or activities for identifying undesirable situations and desirable situations.
- **Problem solving:** Developing a plan with the intent of changing undesirable situations to desirable situations
- **Designing involves both problem understanding and problem solving**
- Designing becomes easier when problem is understood thoroughly

Example: cooking



Undesirable situation: food tasteless

Plan: add adequate salt

Implementation: salt added

Desirable situation: tasty food

Example: Electric sockets



Undesirable situation: open sockets accessible to children is unsafe

Plan: to cover the sockets

Implementation: make and use socket cover

Desirable situation: socket covered and safe

What is Design?

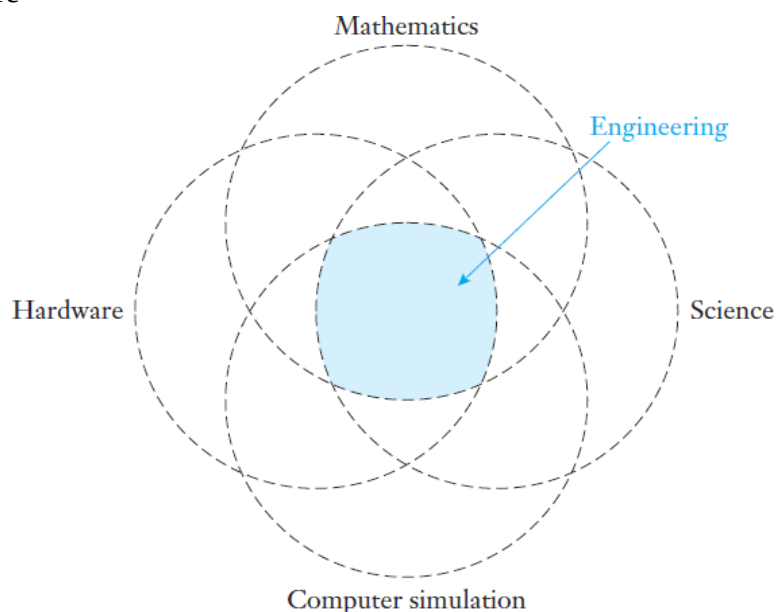
Design: plan of a system, its implementation and utilization for attaining a **goal** (change undesired to desired)

Designing: How a design is developed (**Both Goal and Plan**)

Designs can be for: Technical systems, Educational systems, aesthetic systems (logo design, advertisements), legal systems, social, religious or cultural systems, theories, Models etc.

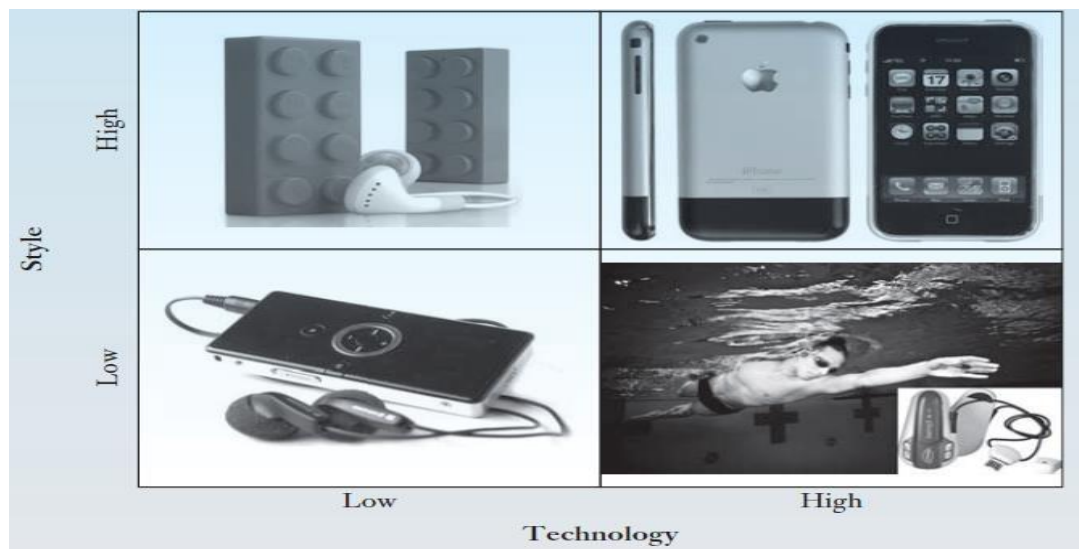
Design Vs Engineering Design

- Engineering is the practical endeavor in which the tools of mathematics and science are applied to develop cost effective solutions to the logical problems facing society.
- Engineers design many of the consumer products **that needed in everyday life**.
- Engineering is all about making useful things that work and impact lives
- The word “Engineering” derives from the Latin root **ingeniere**, meaning to **design or to devise**, which also forms the basis of the word “ingenious
- Engineering is essentially a bridge between scientific discovery and product applications
- Engineers apply their knowledge of **mathematics, science, and materials**—as well as their skills in **communications** and **business**—to develop new and better technologies
- Engineers combine their skills in mathematics, science, computers, and hardware



- “**Engineer creates the new things and makes the old things better and better**”

Style Vs Technology chart for digital music players:



- The above chart shows style on the vertical axis and technology on the horizontal axis.
- This chart provides a frame work to strategically develop innovative products for a wide range of customers.
- In each quadrant is a different digital music players
- **In the lower left:**
The low-style/low-technology is standard affordable player designed for customers who want just to play music. The player, while not the most stylish or high technology, provides solid, expected playback of digital music
- **In the lower right:**
The low style/high tech version is the SwiMP3 player from FINIS. This player integrates water proof technologies with revolutionary bone conduction of sound to provide swimmers with clear sound in under water
- **In the upper left:**
The high style/low tech version is a standard player shaped like a Lego Block, designed for customers who are very style conscious.
- **In the upper right :**
The high style/high tech version is and APPLE i phone for customers who want the latest technologies along with stylish features.

Conclusion:

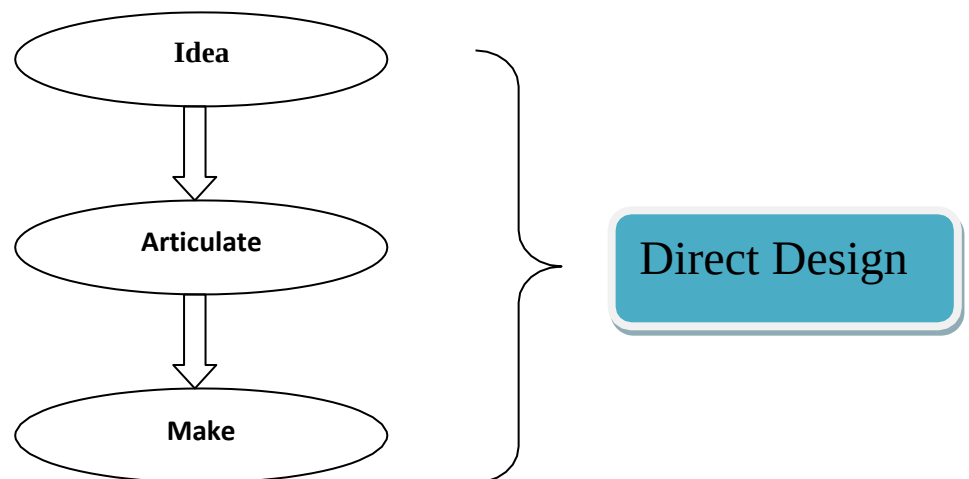
- Developing technically effective, consumer safe, globally aware and environmentally friendly product to meet wide range of social, cultural demands, for that **Engineers can think innovatively**.
- Effective Design through innovation is one of the skills that Engineers will be acquiring during graduation.

❖ Difference between Design and Engineering Design

Design	Engineering Design
The specification of an object manifested by an agent , intended to accomplish goals in a particular environment using a set of primitive components , satisfying a set of requirements subjected to constraints	Engineering Design is the process of devising a system , component or process to meet desired needs. It is decision –making process (often iterative) , in which the basic sciences, mathematics and Engineering sciences are applied to convert resources optimally to meet the stated objective
Design is often used in two different contexts : Action or understanding (verb) Physical construct of object of plan (noun)	The fundamental elements of the Engineering design process include: the establishment of objectives and criteria, synthesis, analysis, construction, testing and evaluation.
There are many ways to define design and they may depend on a specific context or field of design	The formal definition of engineering design depend on the specific engineering

Design to Design thinking

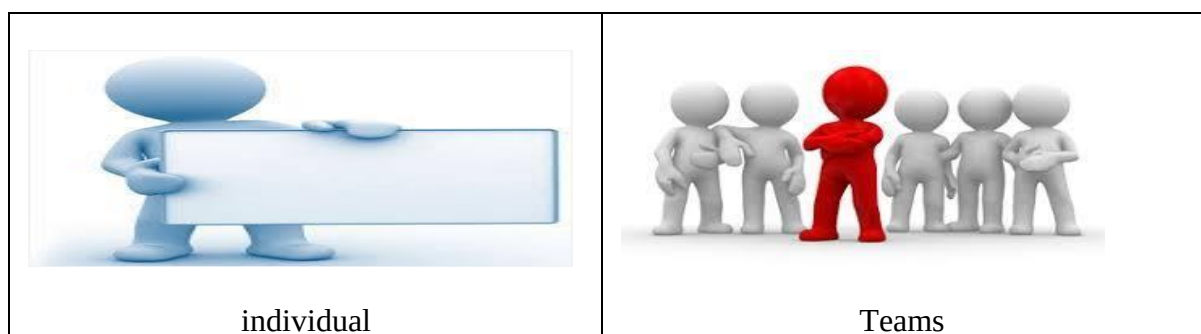
- **Design or making** , has been **classically** understood to be a process of **turning ideas** into **things**
- In this design process there is a vision or an idea, it can be figured out by some drawings work with fabrications of crafts and if everything worked out right then it be materialized.
- So, there is a direct correspondence between ideas, drawings, and finished products. This design process called as **Direct Design**
- This is what most people understand design to be, and what they understand designers to do that they turn their creative ideas into things.

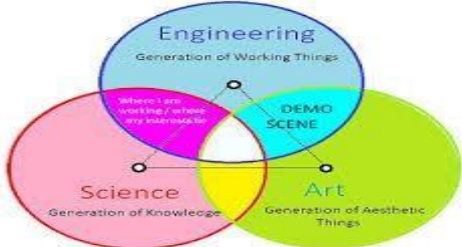
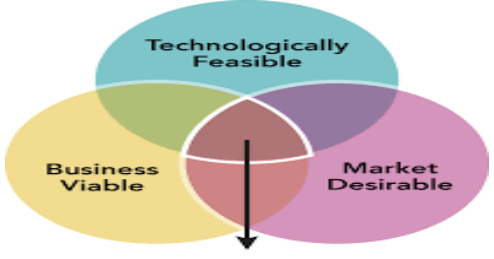


- The primary issue to direct design is preeminence to ideation. The core logic of direct design is that ideas come first and then making comes after.
- While doing the direct design it marginalizes and ignores the **agency of things, environments, users and relations**.
- In 21st century these direct designs were taught as type of closed design process where

designers have ideas and figure out how to realize separately from a deep engagement with the world.

- It becomes obvious that human making could not effectively happen separate from the world of users, practices, problems, needs, politics so on.
- Direct designs are criticized that these are not responsive to real world conditions.
- From an awareness of the power of engagement ,a new and expanded form of design emerged called as **Responsive design**
- **Responsive design** is just that it begins in a considered response of the world rather than an idea comes from designers.
- Responsive design at its best shifted the focus of design way from the narrow idea of designers and design as being focused on independently making beautiful things.
- Design now become about all the **interactive processes** needed to make anything come into being. Responsive design came in many from **environmental design to human centered design**
- In responsive design the most popular form is “**Design thinking**”
- “Design thinking is simply a form of human centered responsive design broad name “Design Thinking”
- The simplest way to understand how responsive design transforms direct design is to see that it adds a new critical step prior to beginning of direct design called **Consultation**. Responsive design does not replace direct design so much subsumes it.
- During consultation the design thinking variants of design asks: what are up to? What are the problems? Then the phase of ideation becomes collective: brain storming, group improvisation and other collaboration exercise are added to mix.
- Then response design works as iterative loop. This loop is significant because it allows the object to evolve through testing and use and not come out of designers thought.
- Response design has had an enormous impact for good. Environmentally centered design is of great value as is user centered design
- A huge part of design thinking appeals that it claims to be an exceptional source of **innovation**



 Products	 Experience
 Design is a combination of Engineering + science + Art	 Design thinking is a combination Technology(Engg) + Business + Humans

Definitions of Design Thinking:

- Design thinking is a methodology that designers use to brainstorm and solve complex problems related to Designing and Design engineering.

Or

- Design thinking is a human- centered approach to innovation that draws from the designer's toolkit to integrate the needs of people , the possibilities of technology, and the requirements for business success—Tim Brown CEO of IDEO

Or

- There is no general agreement on precise definition of design thinking. There are variations across disciplinary cultures, and different meanings depending on its context

Or

- A process that results in a plan of action to improve situation

Or

- An approach that frames problems creatively and generates innovative solutions , strategies , systems and paradigms at the nexus of domain

Or

- Design thinking is a human centered innovation process that emphasizes observations, collaboration, fast learning, visualization and rough prototyping. The objective is to solve not only the stated problems at hand, but the real problems behind the obvious—Thomas Lockwood

Or

- Design thinking refers to creative strategies designers utilize during the process of designing

Or

- Design thinking can be described as a discipline that uses the designer's sensibility and methods to match peoples' needs with what is technologically feasible and what a viable business strategy can convert into customer value and market opportunity---Tim brown

- The term design thinking has been applied in two different approaches related to areas of conceptual development.
- The first established body of knowledge, starting in the late 1960's **investigates how designer** (architects, industrial designers, graphic artists) **perform their craft** and seeks to identify the skills, abilities and knowledge of expert designers.
- The second use of design thinking is related to **business management** .since the mid 1980's interest within the business community grew in exploring how “designer” thinking could be applied to business challenges, and be performed by employees and leaders not trained formally as designers.
- Design thinking is a **blend of logic, powerful imagination, systematic reasoning** and intuition to bring to generate the ideas that consists to solve the problems of the clients with desirable outcomes .it helps to bring creativity with business insights.
- Design thinking helps to **gain a balance between the problem statement and the solution** developed.

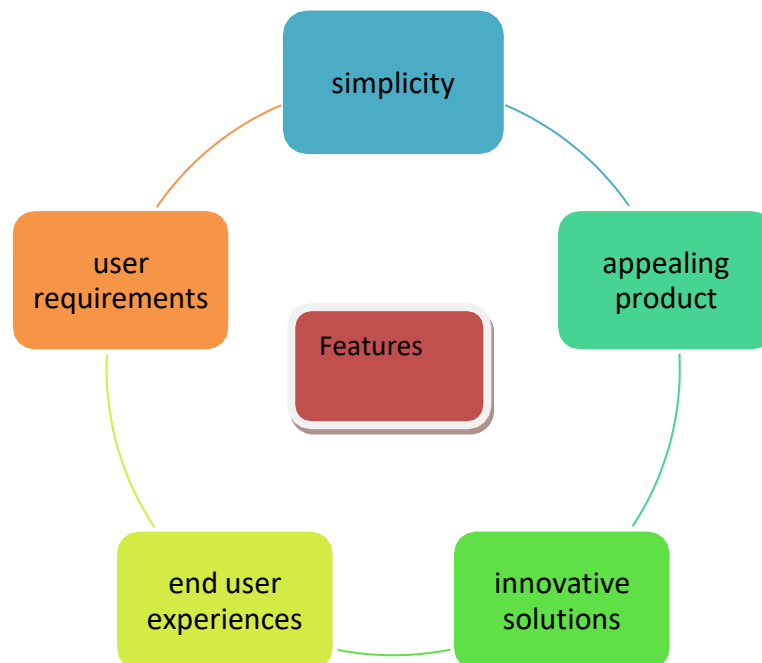


Features of design thinking:

Design thinking provides multi dimensional solutions to the problems.

Features are:

- Finding simplicity in complexities
- Having a beautiful and aesthetically appearing products
- Improving clients and end users quality of experience
- Creating innovative, feasible and viable solution to real world problems.
- Addressing the actual requirements of the end users.

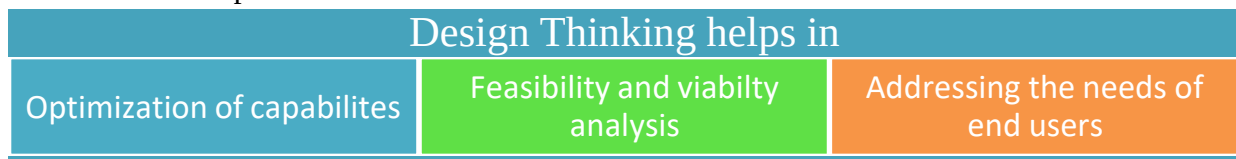


Use of design thinking:

- The basic principle of design thinking is that innovation can be disciplined

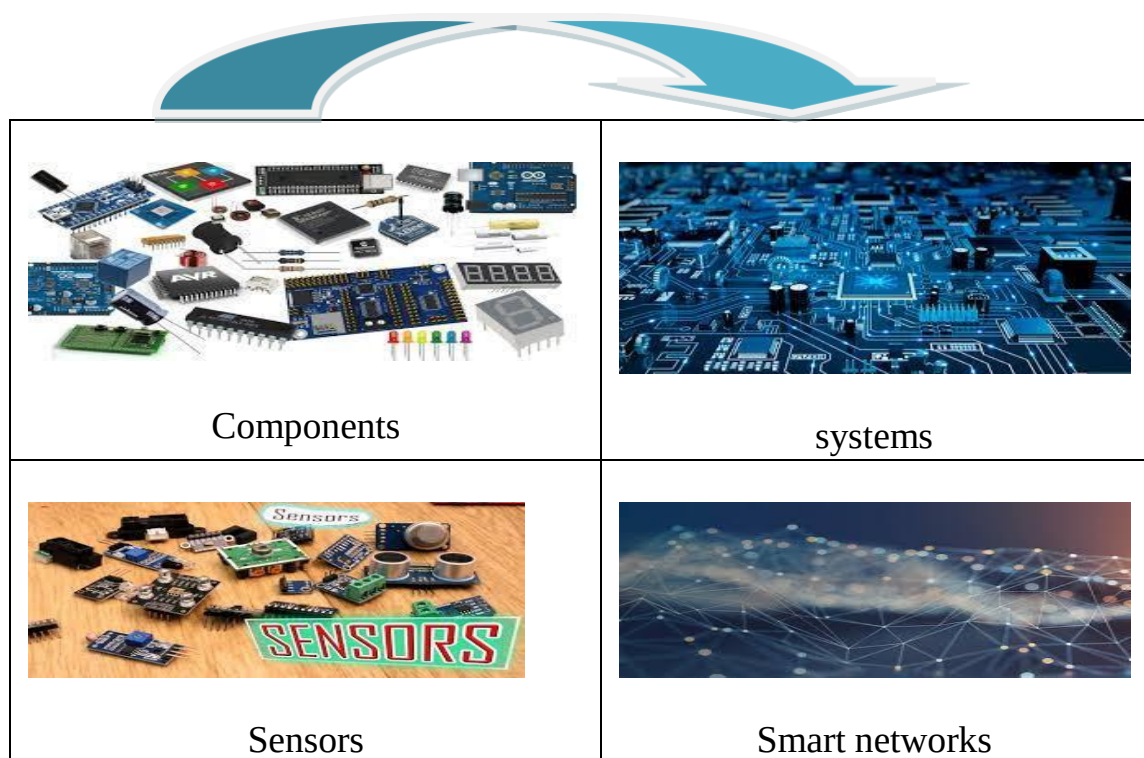
Design thinking helps to learn the following

- How to optimize the ability to innovate
- How to develop a variety of concepts, products, services, processes etc for end-users.
- How to leverage the diverse ideas of innovation
- How to convert useful data , individual insights and vague ideas into feasible reality
- How to connect with the customers and end-users by targeting their actual requirements.
- How to use the different tools used by designers in their profession for solving customers problems



Changing paradigms:

- Changing paradigms can be essential to achieving and attracting what you desire into your life
- A [paradigm](#) by definition is: ' a typical example or pattern of something' or a model. Paradigm is a pattern or programming through constant repetition of a thought, phrase, belief or habit.
- Some paradigms that takes place over the last few decades



 Products	 Services & experiences
 Stand alone system	 Cloud

Conclusion: all these shifts take place not by **Doing the right things but Doing the things right**

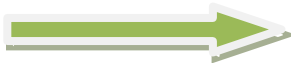
Doing the right things shows path to the problem finding ways

Doing the things right shows path to problem solving ways

Doing the things right be one of the strategy for design thinking

- Design thinking draws upon logic ,imagination, intuition and systemic reasoning , to explore possibilities of what could be and to create desired outcomes that benefit the end user (customer)
- A design mindset is not problem-focused; its solution focused and action oriented. It involves both analysis and imagination.

❖ Thinking of Design



To imaging, visualizing, dream -up, new understanding, new practice and new applications

❖ Thinking about Design



To consider, expectations, capabilities and collaborations

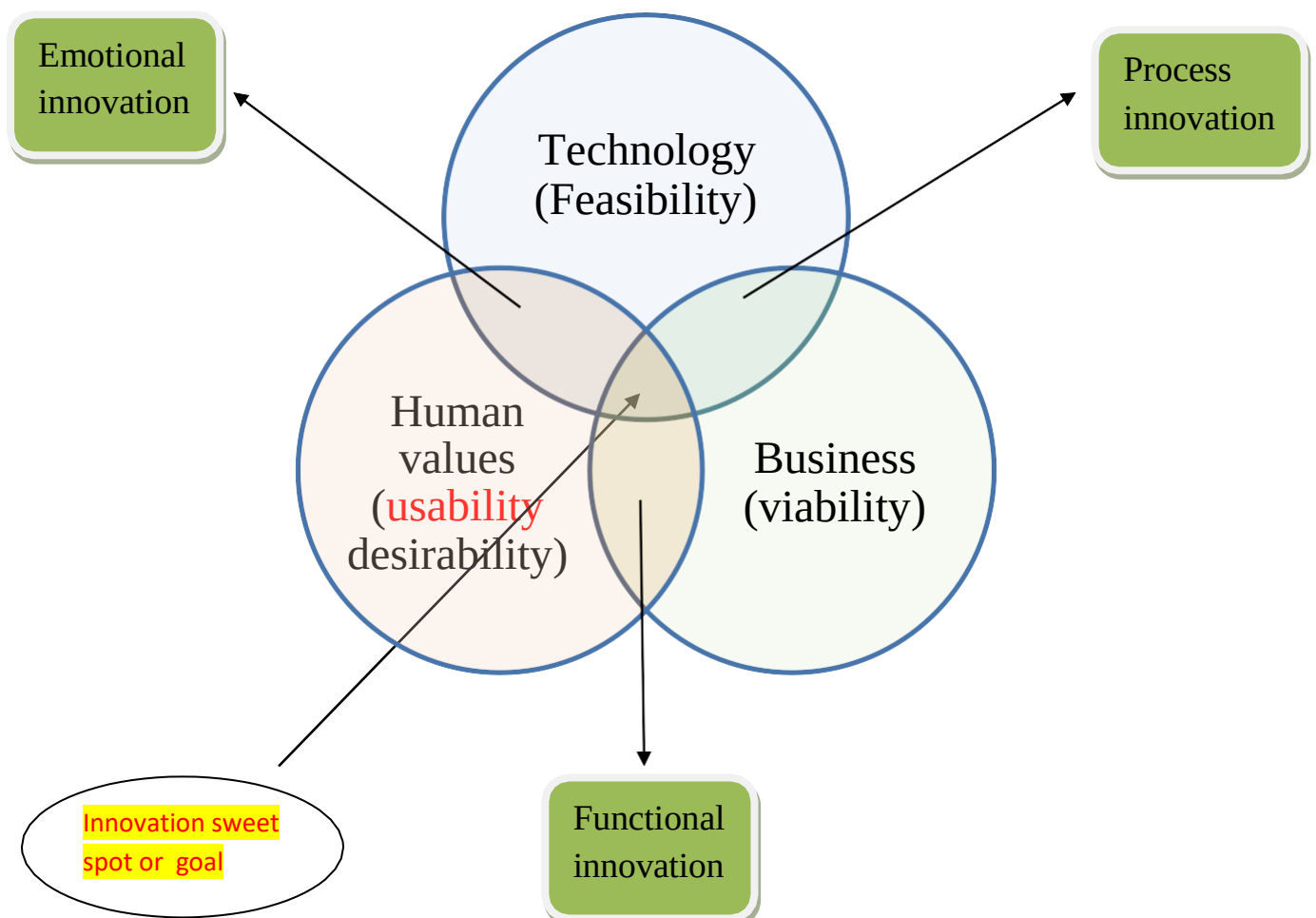
❖ Thinking through design



The ability to use design methods and principles to address uncertainty and complexity

- Desirability tests whether the innovation is solving the customer problem rightly
- Feasibility tests whether the innovation strengthens the business or not
- Viability tests the value chain for long term sustainability

❖ Design thinking & value creation (innovation)



- Desirable ,feasible and viable are three important lenses for innovation through design thinking
- Desirable (people want it) ,feasible (what can actually do it) and it has to be viable(don't go break)

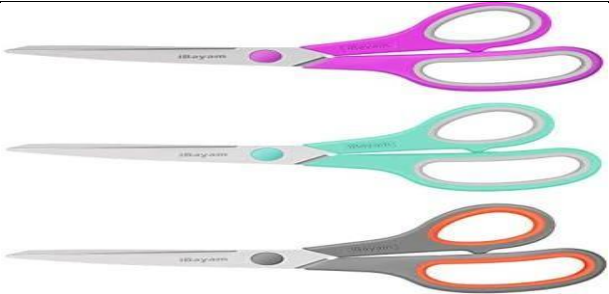
Design thinking resources:

- In order for design thinking to succeed, the right ingredients need to be assembled. The desirable workspace, the materials often used in design thinking, and finally the needed integration and cooperation between the design –thinker's team and the organization.
- The resources are 1. **People** 2. **Place** 3. **Materials** 4. **Organization**

People	1. The individuals that are needed are those who are willing and able to adopt the design –thinking mindset. So the people working on a design thinking are critical to its success 2. These people are experts in system or field or area of the business. They are observant and they listen 3. They have the ability to frame problems and solve them. 4. They can think strategically and execute tactically. 5. They are both creative and analytical 6. They are communicative and comfortable to flexibility 7. The team that emerges from assembling these people should have a spirit of shared purpose, flexibility, collaboration, and mutual support.
Place	1. A space needs to be located that facilitates collaboration and imagination. 2. Design thinking produces many physical artifacts. These artifacts are most useful when they are accessible and visible. 3. Design thinking requires the content under consideration be visible. This translates to the need for stationary and mobile white boards, pin boards display screens, storage buddies and large surfaces on which to hang 4. Flexibility of furniture is essential. 5. According to Lewrick he recommended that 5m² per participant be used as sizing parameter for good space 6. Space, like money and time is often a scarce resource 7. The actual size of the space may become a limiting factor on how large the team and collaborative activities can be.

Materials	1. Making and visual idea transfer are essential elements of design thinking, for that materials are needed 2. list of materials typically needed		
	white boards		

	pin boards	
	whiteboard markers	
	colored pencils	
	sticky notes of various shapes and colors	
	colored adhesive dots	
	Hanging strings	
	colored sheets of paper (A4)	

	pipe cleaners	
	flip charts	
	large rolls of paper	
	tape (cellophane, masking)	
	Glue	
	Lego blocks	
	Scissors	

	index cards	
	audio and video capture tools and analyzers	
	Paper	
	play dough	
	popsicle sticks,	
	foam core	
	Push pins	

	Notepads & pens	
	Tooth picks	
	Laptops	
		
organizations	Locating and accessing design thinking people may require the organization to permit the functional structure to relent and allow enterprise benefit to supersede design thinking objectives	

DIFFERENT KINDS OF THINKING:

ENGINEERING THINKING



Solve your way forward

BUSINESS THINKING



Optimize your way forward

RESEARCH THINKING



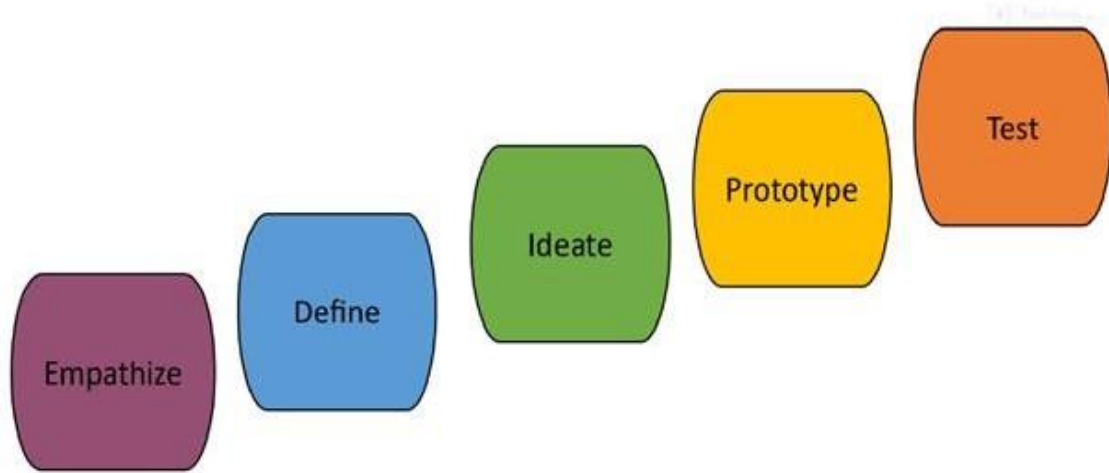
Analyze your way forward

DESIGN THINKING



Build your way forward

- The design thinking process is not necessarily linear nor is there one canonical way to approach it. It is an iterative system with many variations. However, **stand ford's d.school teaches a framework that can help to start the process for almost any problem**



In the Stanford d school's process model, the stages have the following objectives

Empathize	The stage is oriented towards understanding the intended users and the problem from their viewpoint by observation, engagement and immersion.
Define	At this stage the needs and insights discovered in empathize are transformed into an actionable problem statement or design vision tailored for the users.
Ideate	Within the context of the problem statement , the team generates many radical design alternative that explore the solution space
Prototype	At this stage ,promising design alternative are made tangible with which the team, users and others will experience and interact
Test	Prototypes are placed into appropriate contexts of user's lives the goal of gathering thoughtful feedback, learning and refining solutions.

- Beyond suggesting the 5-stage process model, Stanford d.school advocates for six attributes of the design-thinking mindset.

- Those are

- Human centered
- Bias towards action
- Radical collaboration
- Culture of prototyping
- Show, don't tell
- Mindful of process.

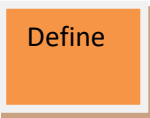
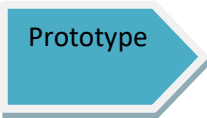
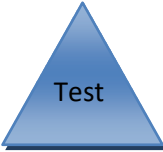
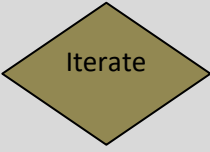
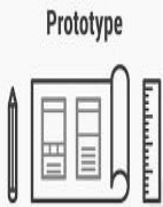
Human centered	It is the idea that design thinkers should identify with the users challenges and develop solutions that address their needs.
Bias towards action	It directs design thinkers to solve problems instead of finding additional ones. Team members should prototype to learn insights instead of expressing self –important personal opinion.
Radical collaboration	It suggests breakthroughs come from a team consisting of members with diverse view points, education and experiences.
Culture of prototyping	It is related to the notion that solutions will arise from trying out concepts and getting users input early and often.
Show ,don't tell	Emphasizes the need to communicate visually rather than verbally
Mindful of process	It encourages teams to be aware of stage the project is in, and what needs to accomplish in the current stage.

- ❖ In addition to the above mind sets views the following points are also considered for good designing thinking process

Those are

- Improve and learn from failure
- Creative confidence
- Growth mindset
- Beginner's mindset

5 stages of Stanford:

1	2	3	4	5	
					
					

Supporting points of Stanford d school frame work.

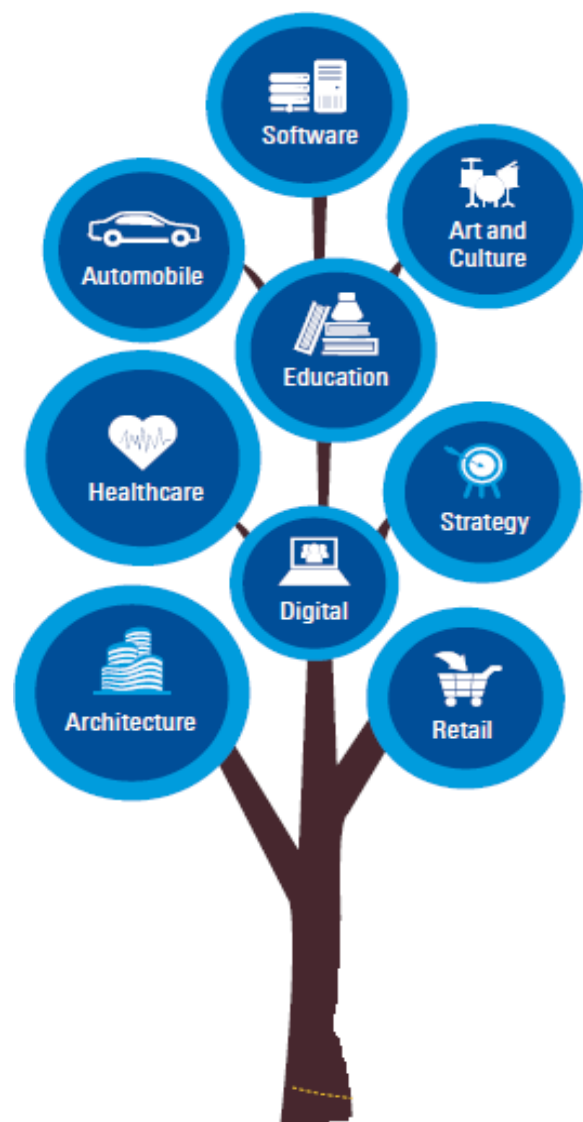
Design thinking is toolkit for creating problem-solving. The **process does not have to be linear**. It can jump from one phase to any other phase based on need.

Application of Design thinking:

- Design thinking is a problem solving that focus on users and their emotional needs while experiencing products and services.
- Design thinking approach is useful for Designers, **Engineers**, planners, Managers, strategists, Economists, Teachers and many other Professions

Design thinking having wide range of applications across different domains:

- Automobile
- Health care
- Architecture
- Software
- Education
- Digital
- Retail
- Art and culture
- strategy



Applications of DT

❖ **Business:**

- Design thinking helps in business by optimizing the process of product creation, marketing, and renewal of contracts.
- All these processes require a companywide focus on the customers and hence, design thinking helps in these processes immensely.
- Design thinking helps the design thinkers to develop deep empathy for their customers and to create solutions that match their needs exactly.

❖ **Information technology:**

- The IT industry makes a lot of products that require trials and proof of concepts.
- The industry needs to empathize with its users and not simply deploy technologies.
- IT is not only about technology or products, but also it is process.
- The developers, analysts, consultants, and managers have to brainstorm on possible ideas for solving the problems of the clients. This is where design thinking helps a lot.

❖ **Education :**

- The education sector can make the best use of design thinking by taking feedback from students on their requirements, goals and challenges they are facing in the classroom.
- By working on their feedback, the instructors come up with solutions to address their challenges.

❖ **Health care:**

- Design thinking helps in health care as well as the expenditure on healthcare. The cost of healthcare facilities is growing day by day.
- Experts worldwide are concerned about how to bring quality healthcare to people at low cost
- Using design thinking, the efficiencies in the system and the perennial crises were addressed

➤ Design thinking for Engineering:

- Design thinking and Engineering systems thinking are two complementary approaches to understanding cognition, organization, and other non technical factors that influence the design and performance of Engineering systems
 - Design thinking methods have been applied to industrial design and product development, while engineering systems thinking is used in professional systems engineering practice and large-scale, complex system design.
 - System dynamics is a foundation method in system engineering – an interdisciplinary field of formalized approaches for designing and managing large-scale, complex engineered systems throughout the life cycle.
 - Systems engineering methodology offers a process for technical management of sophisticated quantitative techniques are used to organize and coordinate work activities, evaluate technical systems interactions, and assure system quality and performance.
 - Systems engineering has drawn from operations research and management science to develop mathematical models of human performance. Example of classical quantitative approaches to management includes Markov analysis, linear /dynamic programming, decision theory, and game theory etc.
 - Systems engineers are well-trained to manage technical system consolidation, but the associated between social and organizational are more difficult to govern in practice.
 - Existing analytical methods hard OR methodology are in applicable or ineffective for solving problems associated between social and organization. Nonetheless ,systems engineering are expected to play ambidextrous roles and have the professional responsibility of “system thinking” i.e **designers who manage technical as well as social and organizational constraints in dynamic environments**
 - Understanding how systems engineers evaluate technical and social relationships- and leverage this information to successfully design and manage complex systems – is a challenge in modern systems engineering practice.

- Engineering systems thinking shares a foundation with systems science, thus values and applications bear strong resemblance to those of general systems theory, cybernetics and systems dynamics.
- Systems receive inputs from the environment, execute processes that transform these inputs into outputs, and send these outputs back into the environment in feedback loop.
- Engineering systems thinking is **system-centered**, i.e it is used for conceptualizing systems, their individual components, and interactions between those components to help designers anticipate emergent features and design robust and resilient systems.
- Contemporary research in **Engineering system thinking seeks** to make the approach **more human-centered** (design thinking approach)

Principles of Design Thinking (Ten Tools/ Commandments)

1 Leave titles at the door!

There is no hierarchy during a Design Thinking workshop. Chef and other rolls are hung on the coat hook.

2 Encourage wild ideas!

Let your imagination run wild. Any (supposedly) crazy idea and every idea should be treated equally.

3 Go for quantity!

Quantity before quality. Selected, analyzed and evaluated later.

4 Build on Ideas of others!

There is no copyright. Ideas from others should be taken up, supplemented or changed.

5 Think human centered!

Design Thinking is first and foremost thinking about people and not about technology or business goals.

6 Be visual and make it tangible!

Use drawings, illustrations, photos, videos, prototypes, etc.

7 Avoid criticism!

Idea generation and evaluation must be strictly separated.

8 Fail early and often!

Failure means learning. Often failure means that you have learned a lot.

9 Stay focused!

Set yourself limits, stick to the concrete tasks in the Design Thinking process**.

10 Let`s have fun!

Developing new ideas in a team should be fun. Creativity needs this fun.

The process of Design Thinking

Phase 1 “Understand“ (Understanding the Problem):

In the first phase it is first about developing an understanding for the challenge/the problem/the need or the requirement (problem understanding). It must be clarified who has to be integrated into the process and, in particular, which technical perspective (process organisation) is necessary. Finally, it must be clarified how the question can best be formulated so that the customer need/problem is defined in concrete terms.

Phase 2 “Observe“:

In this phase, detailed research and on-site observations are carried out on the customer's need/problem. Numerous methods can be used for this, such as interviews, written surveys, observations with recordings through photos or even videos. The results are the clarification of the general conditions, the exact definition of the target group and a comprehensive understanding of the customer and his needs and behaviour.

(*Note :- Phase1 + Phase 2 together forms Empathize age)**

Phase 3 “Point-of-View“ (Define the problem):

After the observations, the findings should next be condensed to a single prototypical user whose problem/need is to be summarized in a clearly defined question.

Phase 4 “Ideate“ (Finding and selecting ideas):

It is only in this phase that the actual brainstorming process takes place. Here the creativity techniques mentioned in chapter 5.3 can be used. Strictly separated from this, the ideas can then be analysed in a customer-oriented manner in order to identify weak points, and a selection decision can be made on the basis of an idea evaluation.

Phase 5 „Prototype“ (Develop the prototype):

In this very important phase, ideas should be visualized as quickly as possible, made tangible, sketched, designed, modelled/simulated, etc. Following the technical field one can speak here of "Rapid Prototyping", whereby the prototype development applies decidedly not only to products, but also to services. A variety of methods for prototype development are available for this purpose.

Phase 6 “Test“:

In this final phase, the ideas are to be further developed and tested through further experiments and customer feedback. In addition, important development, production and market issues have to be clarified.

How to plan a Design Thinking project

- First, the goals for the Design Thinking project are to be defined derived from the company/innovation strategy and the expectations of all participants are to be clarified:
- ☐ Should new ideas be found? If so, for what or for which search field ?
- ☐ Should concrete customer needs and/or certain patterns or trends be found among the customers in a specific subject area?
- ☐ Which goal is to be achieved by when?
- ☐ Which priorities are to be set in terms of content and time for the achievement of the goal?
- ☐ Which employees or which external participants (experts from practice and research, customers, suppliers, external parties from outside the industry) are to be involved from which areas/disciplines?
- ☐ Who will be responsible for project management or moderation?
- ☐ Where are the interfaces between the required disciplines?

☐

Once the goals have been clarified for all those involved, it is certainly necessary to critically review whether the method is at all suitable for the goals.

A decisive success factor for the Design Thinking project is the project organization. In the vast majority of cases, Design Thinking is carried out as a project that involves internal staff as well as external participants from different disciplines. The project team should consist of six to a maximum of nine representatives from different areas (R&D, production, marketing and sales) or disciplines who contribute at least 50% of their working time (and are open to external ideas). Different experience horizons and characters are also helpful to get the diversity necessary for Design Thinking (see chapter 1.1 introductory to the characteristics of Design Thinking).

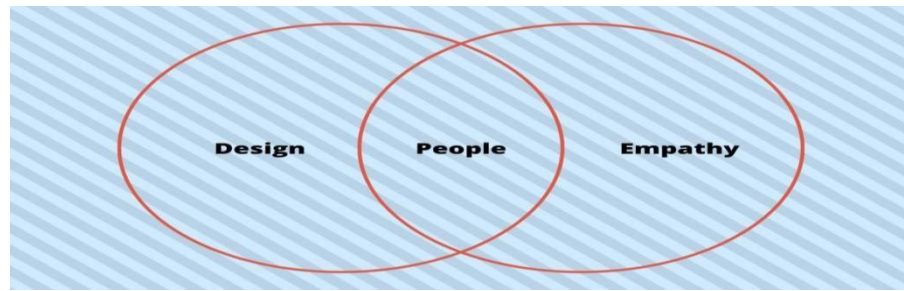
Not to be neglected is also the internal support in the company for the later implementation of the results from Design Thinking. The support of top management, which can be a decisive success factor, is also very helpful here. The support can be seen concretely in the following activities:

- ☐ Adequate resources for the project
- ☐ If applicable, taking over patronage for the Design Thinking project
- ☐ Desire for continuous reporting on the progress of the project
- ☐ Permanent internal and external communication about the strategic importance of Design Thinking
- ☐ Member of the steering committee, if any
- ☐ Personal participation in workshops
- ☐ Recognition of the employees involved

In the case of a more extensive project, a steering committee consisting of managers from the above-mentioned areas must also be set up. Overall, the tasks, competencies and responsibilities must be defined by all project participants.

2.1.1. **Examples of human centered design:**

Human centered approach problem is an approach of problem solving



mostly used in design and management sectors by defining human perspectives in all steps of problem solving.

1. Pull/Push Door:



The Norman Door

Problem	Solution designed
➤ The design had interrupted human brain cognitive Bias. ➤ For brain seeing handle triggers an action to pull it. ➤ In this case the push and pull part of the door are fitted with a handle, this creates confusion in brain.	➤ The push part of the door does not need handle. ➤ Design a door with only in the pull side of the door. ➤ Leave the push side with a plain pad. ➤ People will push the door automatically if they don't see a handle



2. **Kids Toothbrush:**

Kids hold tooth brushes is totally different from an adult. so it is harder for them to use a toothbrush that are designer for adults.



Tooth brush for adult

Problem	Solution designed
➤ Kids hands are so small ➤ They hold tooth brush inside their fist. Adults hold it inside the fingers ➤ Lack of motivation to brush their teeth 	➤ Made the hands of brush fat and squishy ➤ Introduced small bristles ➤ Introduced funny character to the handle ➤ Created funny accessories  Designed by IDEO for Oral-B

3. **Portable Music player –I pod Shuffle**

- Portable music players are introduced in 1997 from that time manufacturers tried to include more and more functionalities of music players. This result in abandoning the user experience of these devices.
- Too much of functionalities and complicated design. The control buttons are too close and accessibility was an issue. Most people use portable music player when they are outside (jogging ,gym, gardening etc) .and it was difficult to hold it during activities

Problem	Solution designed
➤ Inaccessible buttons ➤ Too much features and increased complexity ➤ Difficulty in carrying it 	➤ Simple and accessible buttons ➤ Comes with a clip ➤ Reduced complexity with sleek design  Designed by Apple Inc

4. Ketchup bottle:

- Getting full ketchup out of a ketchup bottle was a pain task. The ketchup was thick and it takes more time to squeeze out till the last drop from that bottle and that's why HEINZ introduced an inverted bottle design.
- HEINZ purchased the inverted bottle design from an American designer named Paul Brown. They redesigned the bottle to have a handgrip and holding space .because of its inverted design , the user will get the last drop from the bottle

Problem	Solution designed
➤ Difficulty in getting full ketchup out of the bottle ➤ Difficulty in handling and squeezing the bottle with one hand 	➤ Designed inverted standing bottle ➤ Designed bottle for easy handling with one hand  Designed by Paul Brown for HEINZ

5. Lay's Stax packaging



- The famous snack and beverage Brand PepsiCo introduce a new type of packaging for lays.
- PepsiCo previous CEO Indra Nooyi Introduced the human centered design approach in lays
- PepsiCo's design team analyzed how people are using their products, and they find out most of the women prefer to eat neat and clean.
- They want to have last chip in that packet but they are not ready to dump all that small cracked chip into their mouth
- Solution is created a vertical and round bottom can with tray, so the user can take out the tray whenever they need a snack and put it back in after use



Problem	Solution designed
➤ Normal package can't be reused ➤ Product wastage is high	➤ Designed around pillar package ➤ Introduce a tray for easy access ➤ Snack size was reduced .so user can finish it in two bites

Winning Companies Use Human Centered Design:

Kellogg's Corn Flakes:



Human centered Design has been around since the 1800's. In 1894, Kellogg Cereal was invented as a more digestible breakfast alternative for hospital patients. In seeing how they struggled to eat toast, Kellogg boiled wheat and rolled the dried flakes to make cereal the patients could eat more easily. He then tried this technique with corn and observed how the patients ate the corn. This became today what we know as Kellogg's Corn Flakes.

This focus on observing human interactions with empathy for their needs is putting the customer at the hub and why human centered design is so valuable to a product's or service's success.

Uvex:



Uvex wanted Altitude to Create a new generation of protective safety eyewear. By incorporating user centered design in their approach to learn what user really want, they were able to design fit logic safety Eyewear , an innovate product design that combines comfortable fit, functionality, and style and subsequently generated high sales for Uvex.

Colgate (Electrical Toothbrushes):



Colgate Hired Altitude to restore their leadership position in electronic toothbrushes. Innovation Strategy and research team determined that users wanted a better fitting, more effective, longer lasting, and easier to maintain toothbrush. They develop the motion, a high-powered, slim profile brush, with dual oppositional oscillating heads and a patented ergonomically correct arcing neck

Role of Empathy in design thinking:

- As the starting point of the design process, Empathy allows a designer to understand the people who will eventually use their product or service
- Empathy is a core value of design thinking .it is also the first step in the design thinking process.
- Empathy, draws attention to the abilities of researchers and designers to see the world through other people's eyes, feel what they feel, and experience things as they do
- Empathy allows a designer to understand the user's physical and emotional needs.
- The Oxford Dictionary defines Empathy as “the ability to understand and share the feeling of another”.
- Empathy is the first step in design thinking because it is a skill that allows us to understand and share the same feelings that others feel. Through empathy, designers are able to put ourselves in other people's shoes and connect with how they might be feeling about their problem, circumstance, or situation

Some questions to consider:

- What is the person feeling?
 - What actions or words indicate this feeling?
 - Can you identify their feelings through words?
 - What words would *you* use to describe their feelings?
- Empathy is the cornerstone of any successful design project. The extent to which you understand and empathize with your users ultimately determines the outcome of your design
 - This means observing and engaging with people in order to understand them on a psychological and emotional level. During this phase, the designer seeks to set aside their assumptions and gather real insights about the user.

- Design Thinking cannot begin without a deeper understanding of the people that designers are designing for. In order to gain those insights, it is important for designer as a design thinker to empathize with the people you are designing for so that you can understand their needs, thoughts, emotions and motivations
- Finally, empathy shows a designer how users think about the world and what is meaningful to users.

2.2.1. **Why Empathy is so important?**

- In a social context, empathy is often what drives us to take action
- If we see people suffering or struggling, and we are able to empathize with their situation, we are compelled to help relieve them in some way.
- Designers need to build empathy for their users in order to take the right course of action
- It's important to understand how the user feels when interacting with a certain product or interface; does the layout of this website evoke feelings of frustration?
- In building empathy, designers can create products which truly please the user and make their lives easier
- Without this empathy, the design process lacks that all-important user-centricity which often marks the distinction between product success and failure.

2.2.2. **Where does empathy fit into the Design Thinking process?**

- During the empathize phase, the designer spends time getting to know the user and understanding their needs, wants, and objectives
- Empathize phase requires you to set aside your assumptions. It's human nature to assume that others will think and feel the same as you in particular situations, but of course this isn't always the case.
- the first step in empathizing with your users is to suspend your own view of the world around you in order to truly see it through your users' eyes
- When it comes to Design Thinking and Human –Centered Design it's time to stop guessing and start gathering real insights about the user!

2.2.3. **What is empathic design?**

- One of the main objectives of empathize stage is to identify user needs and behaviors that are latent, or unarticulated.

- As a designer, it's important to distinguish between what people say they would do in a certain situation, and what they actually do
- In reality, users may have habits or desires that they're not aware of, so it's essential for the designer to observe the user in action
- Empathic research and design is not concerned with facts about the user, such as their age or location. Rather, it focuses on their feelings towards a product and their motivations in certain situations.
- Why do they behave in a certain way? Why do they prefer to do this instead of that? Why do they click here rather than there when presented with a particular screen or page?
- These are the kinds of insights you will uncover during empathize phase, and they will help you to create user experiences that cater to your audience.

2.1. Methods and tools of empathy:

Know The people:

- ❖ The concept of “user-centered” or “human-centered” design is not exactly new
- ❖ In fact, most definitions of design include some reference to crafting things that people use
- ❖ However, as an antidote to the one-size-fits all, mass production age of 19th and 20th century design, it has become important to re-emphasize the centrality of users to the process
- ❖ Design That is sensitive to and based on people's needs and patterns of behavior will be good design.
- ❖ The mode of Know people, with its focus on Empathy, observation, personal engagements, and problem solving, is an indispensable of the design process.
- ❖ Knowing people is about gaining an empathic understanding of thoughts, feelings, and needs by listening, observing, interacting, and analyzing.
- ❖ Immersing yourself in people's daily lives and keenly listening to their stories can reveal very valuable insights, sometimes quite surprising and non-obvious
- ❖ To get to such valuable insights we should focus on everything that people do, say, and think; we should be in the mindset to deeply understand people's activities, needs, motivations, and overall

experiences, just as well as we study our products when we do a product development project.

- ❖ Knowing people well can lead us to entirely new categories of products, services, or business strategies that fundamentally address people's needs and desires, create significant new value, and are very hard to copy.

Methods of Empathy

- Observing Everything
- Building Empathy
- Immersing in daily life
- Listening openly
- Looking for problems and needs

- **Observing Everything:**

- ❖ Observe everything in the context of study, not just the people or the products in use.
- ❖ Notice places notice other people, notice inconsistencies between what people say and what they do.
- ❖ Be prepared to consider innovations that address these seemingly external factors. Even more importantly, look at who and what is not in the field of study

- **Building Empathy:**

- ❖ Is it possible to go beyond just knowing about people's experiences and feelings, to the point of sharing them?
- ❖ e.g Spend a day with a busy mom as she struggles to organize the family's day, share, and identify with her daily experiences, frustrations, and challenges
- ❖ make a deep, direct emotional connection with end users' needs, we will be in a far better position to develop new ideas in tune with the customer

- **Immersing in Daily Life**

- ❖ Spending time with people in their everyday lives can be eye opening. Use the ethnographer's approach to live with and learn about the behaviors, practices, and motivations that form the context in which people will use the tools, artifacts, messages, and

services that you intend to create

- ❖ Spending a day in the life of people for whom you are designing will be revealing.

- **Listening Openly:**

- ❖ We should not just prepare and follow a script for our interaction with research participants. We should let them guide the discussion toward what is important to them; we must be students, not teachers.
- ❖ We ought to think of open-ended questions, suggest general solution alternatives, and be prepared to hear things

- **Looking for Problems and Needs:**

- ❖ What is not working well in the current situation and why? How are people facing challenges in their daily lives? How are they working around the problems? Or are they just giving up since there is nothing that can support their needs?
- ❖ These are great questions that we should ask to reveal opportunities for new products or services

Explore Define phase:

- As a Design Thinker it need to cover all the points and answers that got in the Empathize Phase.
- This is where the process of synthesis comes into picture.
- Clubbing all the answers together and convert them into a **coherent single statement Called Problem statement nothing but Define Phase**
- The first step towards defining a problem is to find who the user is, what is his/her/their needs and then develop insights from the answers
- Think of ‘**How might we?**’ questions
- For example, ‘how might we motivate the employees in DT?’, ‘How might to reduce the cost of knowledge transfer program without compromising its quality and the mandatory pre-requisite resources?’
- The following guidelines will help a design thinker to come up with ‘howmight we’ questions.
 - ❖ **Amplify the good:** A design thinker must think how to amplify the positive aspects of the customers’ needs
 - ❖ **Eliminate the bad:** Design thinkers need to remove all the bad

elements observed in the problem

- ❖ **Explore the opposite:** Design thinkers need to brainstorm on how to convert the problem into an opportunity
- ❖ **Question the Assumptions:** This step involves questioning the assumption at hand
- ❖ **Identify the Unexpected Resources:** Design thinkers should try to find whether some other resources not mentioned by the customer can be leveraged
- ❖ **Create an Analogy:** Design thinking also involves, among many other things, how to create connections between the problem at hand and unrelated images
- ❖ **Break the Problem into Pieces:** This is where again analysis comes into picture for a short while before the problem definition can be synthesized
- Mycoskie navigated the transition from the **Empathize** stage of design thinking to the **Define** stage

- The transition between the **Empathize** and **Define** stages hinges on the concept of **unpacking**
- unpacking tells the talking to, observing and learning about customers and creating a system for turning those insights into something actionable
- Unpacking insights from the Empathize stage does not have to be a complicated process
- The tasks involved in synthesizing empathy and definition can be as simple as breaking out a stack of sticky notes.\
- The primary goal of the unpacking stage is sharing what that learned with other designers and the rest of team members.
- it is important that everyone on the team is on the same page about the information gathered in the Empathize stage.
- Having a complete understanding of the customer and his or her needs will help you form a **problem statement**, which is a written expression of customer's or end user's problem

2.6.1. The Point of View (POV) and its purpose:

- ***Point of view, or POV, in design thinking is a written, actionable statement that expresses the problem that the design team is trying to address. This is also often called a problem statement***
- that is, an expression of the problem (or problems that customers identified the Empathize stage
- In fact, crafting a POV statement is the central purpose of the **Define** stage of design thinking. It provides a framework for designers to use in later brainstorming sessions
- ***“If I had an hour to solve a problem, I’d spend 55 minutes thinking about the problem and five minutes thinking about solutions.” – Albert Einstein***
- **User need statements**, also often called **problem statements** or **point-of-view statements**, are the primary tool in the second stage of design thinking
— the define stage
- A **user need statement** is an actionable problem statement used to summarize who a user is, the user's need, and why the need is important to that user. It defines what you want to solve before you move on to generating potential solutions, in order to 1) condense your perspective on

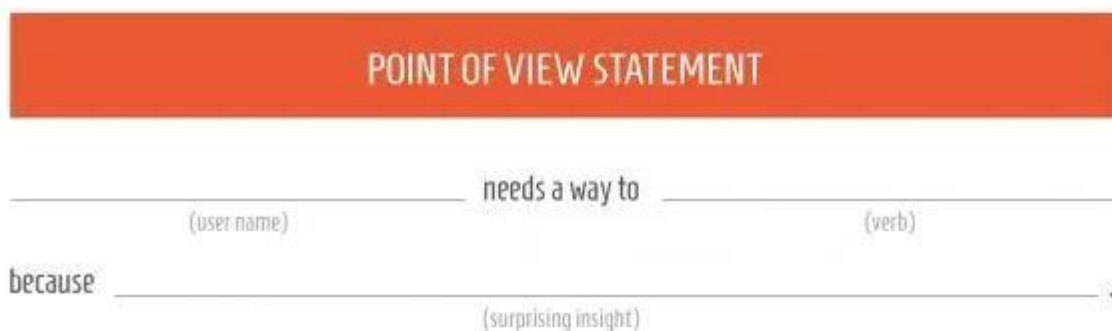
the problem, and 2) provide a metric for success to be used throughout the design thinking process.

How to Create POV:

- ❖ A good POV will allow you to ideate and solve your challenge in a goal-oriented manner – keeping the focus on your users, their needs and your insights about them



- ❖
- ❖ may come up with multiple POVs depending on the insights gained in interviews.



Steps for creating POV:

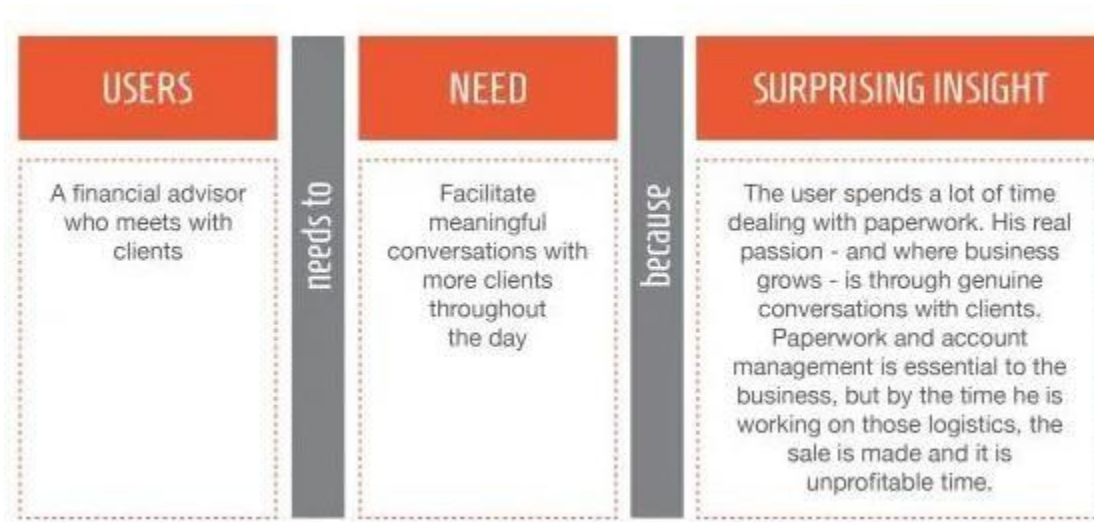
Step 1: Identify Users, Needs & Insights

There are three key components that make up the POV statement:

- **User:** Defines the type of specific user whom writing POV about.
- **Need:** Identifies the users' essential needs/goals.
- **Surprising Insight:** Synthesizes the gathered information into a key takeaway. This statement can be used to design a solution

Step 2: Create Your Chart

Place your scenario into a template like the example below. This creates a guide for structuring your POV statement



Step 3: Form Your POV Statement

This is the fun part where you combine your user, needs and insight into a concise POV statement. If necessary, condense your statements to create a good story.

More Tips for Your POV Statements

- Keep the focus narrow.
- Frame the challenge as a problem statement.
- Use them to evaluate competing ideas.
- Make them actionable.
- Remember, these statements will guide your innovation efforts and provide inspiration for your team

❖ *Note: Since feelings, thoughts, and beliefs cannot be directly observed, infer those by carefully paying attention to clues*

- **Write down needs.** “Needs” are emotional or physical necessities. They are activities and desires, *not* solutions. Identify needs from traits or from contradictions – such as a disconnect between what he/she says and does. Write them down on the side of your empathy map
- **Write down insights.** An “insight” is a remarkable realization that you could leverage to better respond to a challenge. Insights often grow from contradictions between two user attributes (either within a quadrant or from two different quadrants) or from asking yourself “Why?” when you notice strange behavior. Write down potential insights on the side of your map. Capture tensions and contradictions as you work.

What is the Define stage of Design Thinking?

Imagine your team meets together to solve the biggest marketing issue. However, someone has neglected to define what the biggest issue is, so your marketing team arrives with solutions to email marketing processes, social media marketing concepting problems, and ways to fix the quality of coffee in the break room.

The Define stage of Design Thinking first identifies the problem designers are trying to solve. This keeps everyone oriented to the same solution. This stage also helps to define the problem in the most beneficial way: it should be broad but not too obscure and narrow but not too limiting. It’s best if you can distill your problem into a single statement.

4 fundamental questions for the Define stage

Examining the problem from multiple angles is the best way to understand the core issue at play. But that’s also a daunting task that can feel too obscure to be helpful. Luckily, there are guideposts to help you get started. By answering a few fundamental questions, you can formulate a better definition of your problem.

1) Who’s having the problem? This is your core user. Start by defining your target user, their desires and motivations, and how they interact with your product. Without knowing who you’re trying to help, you’ll be unable to actually deliver value to their life.

2) What problem is your user actually having? If you’re designing a car buying platform, you may think the problem you’re trying to solve is how to offer a greater array of car buying options. But your core user may not actually be suffering from options, but rather from indecision. Examine the pain points you identified during the Empathize stage and determine what the user really needs. Then you can also brainstorm different ways to solve this problem.

3) Where is the issue? This is important to UX designers because the issue

may only be in one specific area (i.e. the mobile app or the desktop version or within one portion of the product). This is a great step because it allows you to hone in your focus on one specific space. Or, if the problem presents in multiple spaces, you'll better understand the contexts in which it must be remedied.

4) Why? This question is perhaps the most profound of all four fundamental questions. It asks what it would mean to your user if the problem were solved. What value would be gained to the user? On a larger scale, how would solving the user's problem impact the entire business?

Best practices for the Define stage

Just as there are questions to help you complete the Define stage, there are also several best practices you should follow as you embrace this step of Design Thinking.

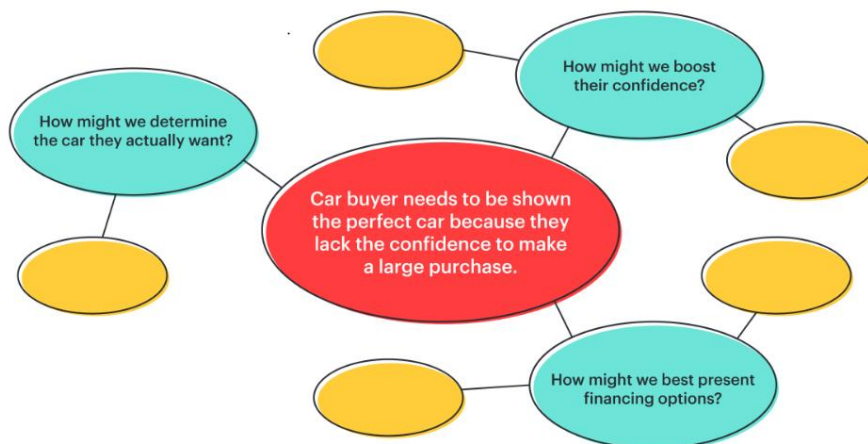
Use a Point of View (POV) statement

This single statement is the summation of your work. It defines who your user is, what their needs are, and any surprising elements or insights you've gathered from your research. This point of statement can follow a formula: (user) needs to (verb) because (surprising element or insight)

So, in our car scenario above, the point of view statement could look like this: "Brian, the Indecisive Car Buyer, needs to be shown the perfect car for him because lacks the confidence to make a large purchase." Keep your POV statement centered on the user.

Ask "how might we?" (HMW)

Once you have your POV statement, you can determine opportunities for solving the user's problem within design. Examine your POV statement and brainstorm topics that stem from the problem. Then turn those subtopics and issues into a question by adding "How might we..." before them. For instance, using the above POV statement, a few "How might we" questions could be:



How might we determine the car Brian actually wants?

How might we boost Brian's confidence?

How might we best present financing options?

Your “How might we” statements should allow you to come up with many possible solutions, including solutions that seem outlandish or not feasible. The point here is to generate a pool of solutions so that you can pick solutions that seem especially valid and prioritize them.

Use the 5 Whys

The 5 Whys technique was developed by Toyota as part of its problem-solving training to get to the root cause of a problem. It involves asking an initial question and then asking “why” for each subsequent answer. So, with our Brian example, it could look like this:

Why does Brian lack confidence about buying a new car? Because he’s overwhelmed by the options.

Why do the options seem overwhelming to him? Because he doesn’t think he knows enough about cars to make a good decision.

Why does he think he doesn’t know enough about cars? Because when he goes car shopping, he doesn’t know what questions to ask.

Why doesn’t he know what questions to ask? Because he doesn’t know what’s most important to him in a car.

Why doesn’t he know what’s most important to him in a car? Because he’s never been forced to think about it.

So, the root issue of Brian’s lack of confidence for purchasing a new car could be that he doesn’t know what he actually wants. One solution could be a quiz in your product that helps Brian determine his top priorities in a car.

Why/how laddering

This technique is used to discover a variety of user needs and actions that can be taken to meet those needs. The ladder begins by asking a fundamental question, then asking why several times until a more abstract statement emerges with core feelings from the user. Then start with to work back down that abstract statement by asking “how” about each statement. You should end up with a hierarchy of your user’s needs that can help you better formulate a variety of solutions.

Collect and unpack user stories

By unpacking user stories in a user story map, you can better determine which tasks and activities hold the most discomfort for your user, and then formulate solutions to that discomfort.

If you know, for instance, that Brian hates logging into websites, you can consider alternative solutions to creating a password and username. Visualizing a user story map will help you see where opportunities lie.

**Question Bank:
(Short Answers)**

Q.No.	Question	Reference page number (Above)for answer
1	What do you mean by Discovery, Invention and Innovation?	11, 12
2	Define Design Thinking?	11, 12
3	List 5 stages of Design thinking?	23
4	What are the 3 key elements of Design Thinking & value creation?	16
5	List 4 resources of Design Thinking?	17
6	Give any 4 applications of Design Thinking?	36
7	List 5 methods of Empathy?	42
8	What do you mean by POV (Point of View)?	45
9	List 3 steps involved in PV (Point of View)?	46
10	What is the meaning of HMW?	49

**Question Bank:
(Descriptive Answers)**

Q.No.	Question	Reference page number (Above)for answer
1	Explain the stage of “Empathize” in Design Thinking? Apply the “ Empathize “ stage to any one real life product or service?	30
2	What are the 10 principles/ tools/ commandments of Design Thinking?	29
3	Apply Design Thinking to identify the problem and propose a solution to any 2 real life application?	37, 38
4	Explain the role of Empathy ind Design Thinking?	39, 40
5	Why Empathy is so important in Design thinking?	40
6	Explain 4 fundamental questions of Define stage ?	48, 49